

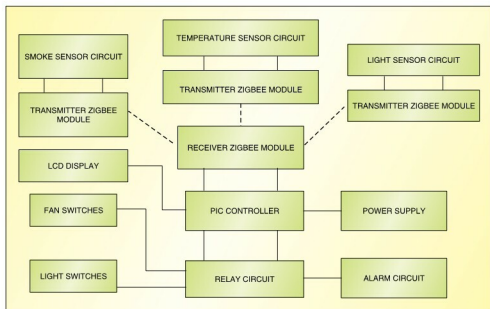


Fig. 3: Block diagram of the proposed home automation network

AC turn off.

Display unit. All the activities that are taking place with ZigBee device, sensors and controller are shown on the LCD panel.

Other components

Controller. PIC 18F4522 microcontroller is proposed for implementation of the home automation wireless sensor network to control the operation. Other families of microcontroller units can also be used depending upon the use and complication of the automation network. The microcontroller controls the network formed by ZigBee. The signals sensed and transmitted by the coordinator unit (which is attached to the controller unit) are guided to the proper relay circuit. The switching of the relay is also done by the controller unit.

Temperature sensor. Various types of temperature sensors are available in the market so sensors suitable for the requirement can be used. The LM358 is a commonly used temperature sensor. It has temperature range of -55° to $+150^{\circ}\text{C}$ and can be used with single power supply or dual (plus and minus) power supplies. The output voltage of LM358 is linearly proportional to the Cel-

sus (Centigrade) temperature. The LM35 thus has an advantage over linear temperature sensors calibrated in $^{\circ}\text{Kelvin}$, as the user is not required to convert the reading to obtain convenient Centigrade scaling. Low output impedance, linear output and precise inherent calibration make interfacing of the LM358 to the readout or control circuitry easy.

Light sensor. A light sensor is required to activate lights in the house. A simple light-dependent resistor (LDR) with proper light arrangement can be used to operate as the light sensor. It is a cost-effective option for use in the home automation sensor network.

A light-dependent resistor, also known as photo-resistor or cadmium-sulfide cell, is a resistor whose resistance decreases with increase in the incident light intensity. It can also be referred to as a photoconductor. A photo-resistor is made of a high-resistance semiconductor. If light falling on the device has a high enough frequency, photons absorbed by the semiconductor give bound electrons enough energy to jump into the conduction band. The resulting free electrons conduct electricity, thereby lowering resistance. A photoelectric device can be either intrinsic or extrinsic.

Smoke sensor. Smoke sensors are available in the market with a number of features. For detection of the smoke or fire, you can implement a smoke sensor with the help of an optocoupler sensor. An optocoupler is a device that uses a short optical transmission path to transfer an electronic signal between the elements of a circuit, typically a transmitter and a receiver, while keeping them electrically isolated—since the electrical signal is converted into a light beam, transferred, then converted back into an electrical signal, there is no need for electrical connection between the source and

destination circuits.

The opto-isolator is simply a package that contains both an infrared light-emitting diode (LED) and a photo-detector such as a photosensitive silicon diode, Darlington transistor pair or silicon-controlled rectifier. Wavelength responses of the two devices are tailored to be as identical as possible to permit the highest measure of coupling. If this configuration senses any smoke or fire, the alarm sounds and the water sprinkler system is turned on.

Comparison with other alternatives

With the ZigBee network, you can do away with the complex wiring of wired automation. ZigBee-based home automation provides much higher operating range than Bluetooth and saves considerable amount of power. Also, as it is flexible and compatible with future technologies, it can be easily customised for individual requirements. ■

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